

Lesson I

What is an OS?

- An OS is a software that connects the hardware and software components to perform operations within the computer.
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Why do we need an OS?

- Convenience for users
 - Efficient use of hardware
 - Resource management
 - Program execution
 - System security
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User convenient / friendly OS

- Provides GUI
 - Provides CLI
 - Easy interaction with systems
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OS resources

- Processor(CPU)
- Main Memory(RAM)

- Secondary Memory(HDD/SSD)
- Input Devices
- Output Devices

RAM vs ROM

Difference Between RAM and ROM

RAM	ROM
Volatile memory (data lost when power off)	Non-volatile memory (data saved permanently)
High speed, supports fast read and write	Slower compared to RAM
Larger, measured in GBs (e.g., 8GB, 16GB)	Smaller, measured in MBs (e.g., 4MB, 8MB)
Example include: DRAM, SRAM	Example include: PROM, EPROM, EEPROM
Easily upgradable (add more RAM sticks)	Not usually upgradable (built into motherboard)

HDD vs SSD

	SSD vs. HDD Usually 10,000 or 15,000 rpm SAS drives	
0.1 ms	Access Times SSDs exhibit virtually no access time	5.5-8.0 ms
SSDs deliver at least 6000 io/s	Random I/O Performance SSDs are at least 15 times faster than HDDs	HDDs reach up to 400 io/s
SSDs have a failure rate of less than 0.5%	Reliability This makes SSDs 4-10 times more reliable	HDDs failure rate fluctuates between 2-5%
SSDs consume between 2 and 5 watts	Energy Savings This means that on a large server, approximately 100 watts are saved	HDDs consume between 6 and 15 watts
SSDs have an average I/O wait of 1%	CPU Power You will have an extra 6% of CPU power for other operations	HDDs average I/O wait is about 7%
The average service time for an I/O request while running a backup remain below 20 ms	Input/Output Request Times SSDs allow for much faster data access	The I/O request time with HDDs during backup rises up to 400-500 ms
SSD backups take about 6 hours	Backup Rates SSDs allow for 3-5 times faster backup for your data	HDD backups take up to 20-24 hours

Core functions of an OS

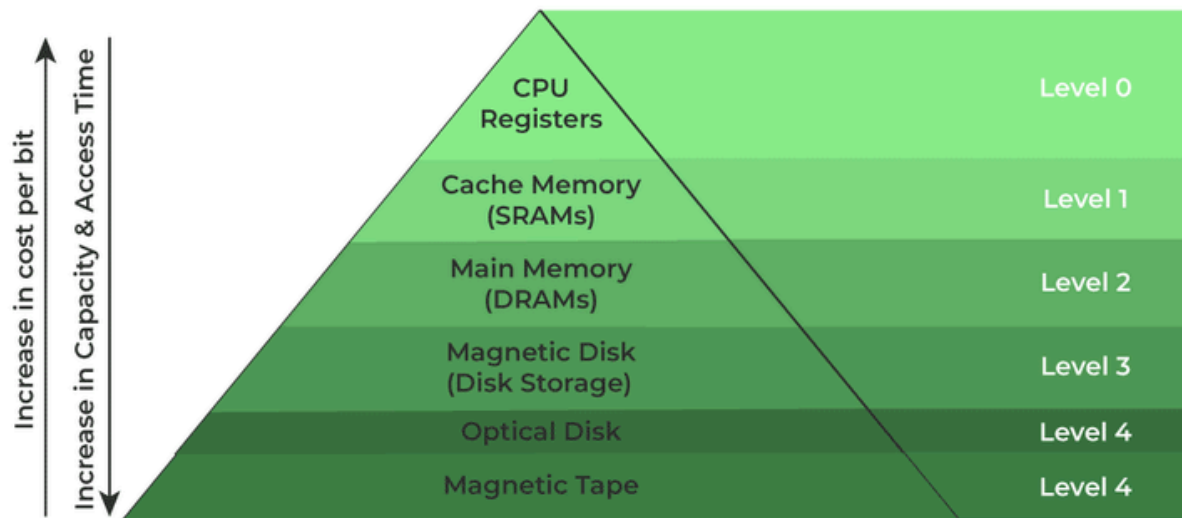
- Control Hardware resources
- Provide services to Applications

Memory Management

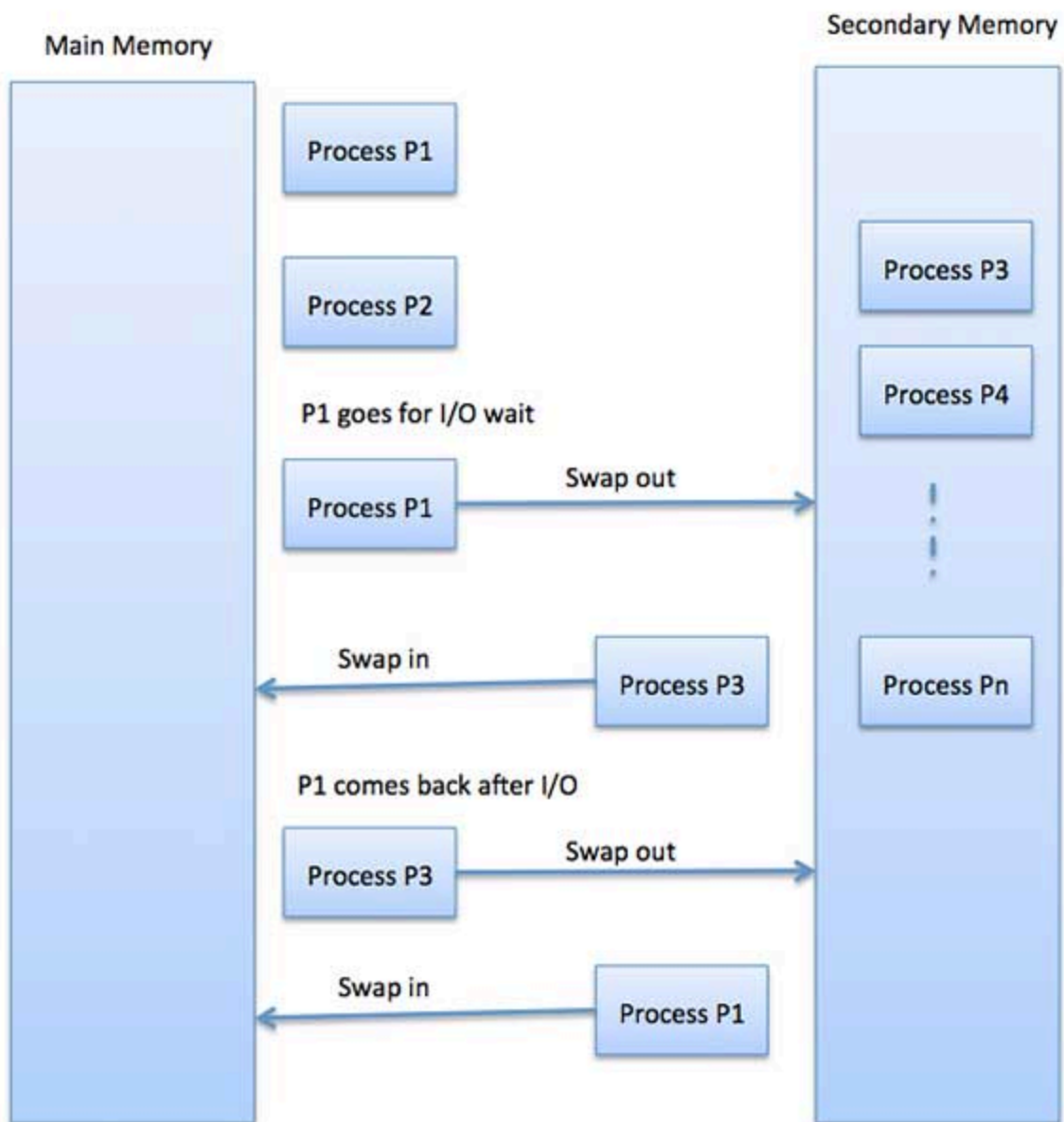
- Allocating memory for programs.
- Deallocate memory after use.
- Tracking memory use.

Key Functions

- Keeps track of which memory locations are free and which are currently occupied.
- Allocate free memory to a process when needed.
- Protect one process's memory from another.
- Supports virtual memory(Using secondary memory as replacement RAM)
Ex: SWAP memory



Memory Hierarchy Design



Device Management

- Controls all I/O devices
- OS communicate with devices using device drivers

Key Functions

- Allocate devices to processes
- Manage device drivers
- Handling buffering, caching and spooling
- Resolve device conflicts

Process Management

- Process Management handles;
CPU allocation to processes
Process execution and termination
Scheduling of tasks

Key Functions

- Create or delete processes
- Schedule processes using algorithms
- Perform context switching
- Support multitasking

Ex:

5 Stages of Process Management

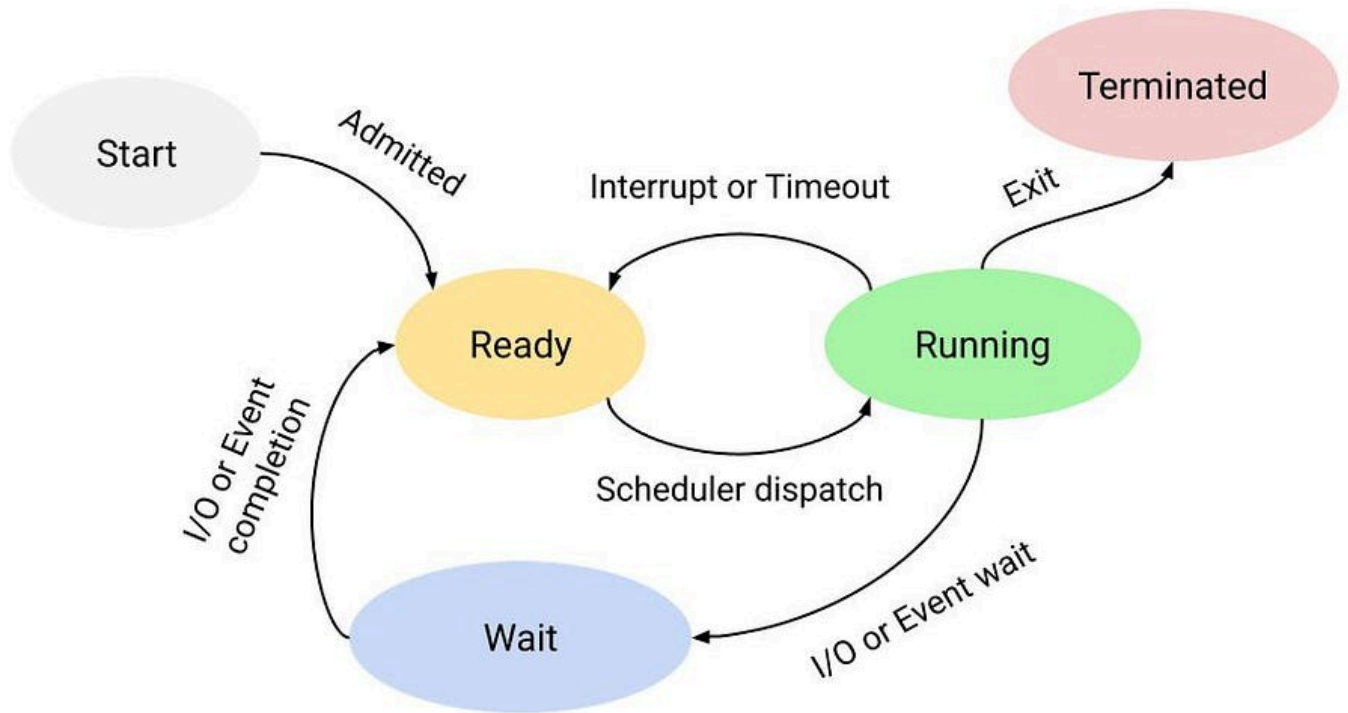
1. Start

2. Ready

3. Running

4. Wait

5. Terminate



File Management

- Organize and control storage on disks
- Files are logical units of information stored permanently
- Creation and deletion of files
- Directory management
- File access permissions
- Read/Write/Update Operations

3 File management software

- File naming
- Batch document scanning
- Folder organization
- Robust reporting
- Quick searching

- Network/cloud file sharing
- Administrative tools

Security Management

- Security protects system resources from;
 1. Unauthorized access
 2. Malicious attacks
 3. Accidental damage
- Essential for multi-user systems
- User authentication(username & password)
- Authentication and access control
- Data protection and encryption
- Auditing and logging

Error detecting and error handling

- Errors occur due to;
 1. Hardware failures
 2. Software bugs
 3. User mistakes
- OS must ensure system stability
- Detect CPU, memory, I/O errors
- Logs error details
- Take corrective action

Co-ordination between Users, Software and Hardware

- OS as a coordinator
- OS acts as an intermediary between;
 1. Users
 2. Application software

3. Hardware

Job accounting / scheduling

- Job accounting tracks;
 1. Resource usage by user and processes
 2. CPU time, memory usage, I/O operations
- Important for **System Monitoring and billing**.
- Maintain logs of resource usage
- Helps in performance analysis
- Support fair resource allocation
- Useful in multi-user

Context Switching